HIMANSHU SHUKLA

MACHINE LEARNING ENGINEERING / APPLIED AI

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PROFESSIONAL PROFILE

Mathematics & Computing undergraduate working on cross-modal image retrieval, embedding search, evaluation contracts and computer-vision prototypes. Focused on measurable experiments and explicitly stated validation limits.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior | B.Tech Mathematics & Computing | Expected 2028 | CGPA 7.75/10 (through Semester IV)

TECHNICAL SKILLS

ML & retrieval: PyTorch, scikit-learn, NumPy, Pandas, contrastive learning, embeddings, FAISS, F1@K

Programming: Python, C++, MATLAB, SQL, PostgreSQL

Experiments & tools: FastAPI, pytest, Docker, Jupyter, Git/GitHub, manifest validation, latency measurement

SELECTED PROJECTS

EarthBridge | Cross-modal satellite-image retrieval | PyTorch, FAISS

- Developed a research prototype supporting optical, SAR and multispectral descriptor retrieval with contrastive representations and a FastAPI interface.
- Built evaluation procedures for F1@5/F1@10, leakage-aware data partitioning, latency checks and a 128-pair overfitting gate; exact performance claims require the associated experiment record.

IrisTrack | Opt-in webcam tracking prototype | Computer vision

- Developed face/iris landmarks, blink and head-pose quality checks, calibration and validation flows; physical webcam accuracy and installer readiness remain unverified.

Mulyankan | Evidence-linked evaluation | React, TypeScript, Supabase

- Designed evidence references, insufficient-evidence flags and human reviewer controls; demonstrated the research-evaluation prototype at CMPDI in July 2026.

Sentinel | Flow-monitoring prototype | Python, Scapy, FastAPI

- Created local packet/flow monitoring views and a preliminary analysis interface, without claiming production intrusion-detection performance.

ADDITIONAL WORK

NewsSources (PDF/article ingestion, provenance); OHS (React/Firebase real-time collaboration); Mulyankan Spatial Lab (3D explanatory interface).

LEADERSHIP & RECOGNITION

Class Representative (3rd Year); NPTEL Coordinator; Training & Placement Cell member; SchoolBuddies co-founder. First place — Neeti Summit English Debate.